

Assignment 06

Third Year BS(Honours) 2024-2025

Course Title: Math Lab III Course Code: AMTH 350

Department of Applied Mathematics, University of Dhaka

Name:

Roll:

Group:

Day 1

1. A stock price is currently \$40. Over each of the next three 2-month periods it is expected to go up by 10% or down by 10%. The risk-free interest rate is 12% per annum.
 - a. What is the value of a 6-month European call option with a strike price of \$42. If the call were American, would it ever be worth exercising it early.
 - b. Using Binomial tree find the final value of a 6-month European put option with a strike price of \$42.
 - c. Verify that European call and European put prices satisfy put-call parity.
2. Consider a European call option on a non-dividend paying stock where the stock price is \$40, the strike price is \$40, the risk-free rate is 4% per annum, the volatility is 30% per annum, and the time of maturity 6 months.
 - a. Find the value of the option using two-step tree. If the call were American, would it ever be worth exercising it early.
 - b. Find the value of the European call option with 50, and 100 time steps. Also using put-call parity find the European put option. Show these results in a table with appropriate heading.
3. Suppose you have the choice of investing \$1000 in just one of two ways. Each investment will pay you an amount listed in the table below at the end of the next five years.

Year	1	2	3	4	5
Investment A	225	215	250	225	205
Investment B	220	225	250	250	210

- a. Using the present value of the investment to make the decision, which investment would you choose? Assume the simple annual interest rate is 4.33%.
 - b. Using the rate of return per year of the investment to make the decision, which investment would you choose?
4. Suppose that a stock price process for discrete time interval is approximated as

$$S(t + \Delta t) - S(t) = S(t)\mu\Delta t + \sigma S(t)\epsilon\sqrt{\Delta t}$$

where, $S(t)$ is the value of stock at time t , μ is the expected return in the risk neutral world, σ is the volatility, ϵ is a random variable that follow standard normal distribution with mean 0 and standard deviation 1, and Δt is the time interval. The stock price has an expected return of 16% per annum and a volatility of 30% per annum. The initial stock price is \$100. Construct 10 random paths for the stock over 2 months using **Monte Carlo simulation**. Show the path with appropriate axes labels. Hence, find a two-month European Call option price when the strike price is \$95.

Day 2

5. Repeat question 4 to find two month European call option price for increasing number of random paths (N) with $N = 50, 100, 200, 300$. Plot the result with appropriate heading and axes labels and draw a vertical line for exact values to show how the results converges with increasing random path. Also compare your results with exact value calculated from pricing formula by constructing a table with the heading as follows:

“N”, “Approximated Value” “Exact Value” “Absolute Error”

6. Plot the function $f(x) = \begin{cases} \sin x, & |x| < \pi \\ 0, & \text{Otherwise} \end{cases}$ and find its Fourier transformation $\hat{f}$. If $\hat{f}$ is real valued, plot it; otherwise plot $|\hat{f}|$.
7. Let the complex potential function be defined by $f(z) = z^2$ where $z = x + iy$
- Express $f(z)$ in the form $f(z) = \phi(x, y) + i\psi(x, y)$ and determine whether the function $f(z)$ is analytic in the complex plane. Justify your answer using Cauchy-Riemann equations.
 - Show that both $\phi(x, y)$ and $\psi(x, y)$ are harmonic functions.
 - If $\phi(x, y)$ velocity potential and $\psi(x, y)$ is stream function, determine the velocity field components.
 - Using MATLAB to plot streamlines, equipotential lines, velocity vector field and also comment on the symmetry and stagnation points of flow.
8. Consider the complex valued function
- $$f(z) = \frac{z+1}{z(z-2)(z+i)} \text{ where } z = x + iy$$
- Identify all singular points of $f(z)$ in the complex plane and classify each singularity.
 - Using MATLAB, generate a contour plot of the magnitude $|f(z)|$ over an appropriate region of the complex plane. On the same figure: mark the singular points clearly, label each singularity, and show the real and imaginary axes.
 - Compute the residues of $f(z)$ at each singularity.
 - Evaluate the contour integral $\oint_C f(z)dz$, using Cauchy Residue theorem, where C is the contour as (i) $|z| = 2$, and (ii) $|z| = 1.5$. For each case, identify the poles enclosed by the contour and visualize the poles and contours using MATLAB.